

Dataset Links

All sources are free. Every link below is drawn from the project brief and the master dataset reference. Click any blue label to open; the raw URL is shown beneath it for copy-paste.

CORE SOURCES — PRICES, MACRO & THE COVID OVERLAY

1 · Sector ETFs + SPY daily OHLCV

Primary price panel — returns, volatility, drawdowns, recovery, correlations, abnormal returns. Fetched via a library, never a raw endpoint. Universe: XLK, XLF, XLE, XLV, XLY, XLP, XLI, XLU, XLB, XLRE, XLC + SPY.

[yfinance library \(GitHub\)](#)

<https://github.com/ranaroussi/yfinance>

[pandas-datareader — StoogDailyReader \(docs\)](#)

<https://pandas-datareader.readthedocs.io/en/latest/readers/stoog.html>

⚠ Stoog's *direct* CSV endpoint is behind a JS anti-bot wall — use the library fetchers (yfinance or pandas-datareader), hit the network once, and cache to Parquet. yfinance is fragile (HTTP 429) → retry / back-off.

2 · FRED — Federal Reserve Economic Data

Macro-linkage + VIX beta. Series used: VIXCLS (daily), UNRATE, INDPRO, CPIAUCSL, GDP, and sector payrolls MANEMP / USCONS / USTRATE.

[Request a free API key \(32-char\)](#)

<https://fredaccount.stlouisfed.org/apikeys>

[FRED API documentation](#)

<https://fred.stlouisfed.org/docs/api/fred/>

[Example series page — VIXCLS \(CBOE Volatility Index\)](#)

<https://fred.stlouisfed.org/series/VIXCLS>

› Use the fredapi package or `pandas_datareader.DataReader(name, 'fred')`. ~120 requests/min.

3 · Our World in Data — COVID-19

Stringency index + case / death waves overlay on sector returns.

[Full dataset — direct CSV](#)

<https://covid.ourworldindata.org/data/owid-covid-data.csv>

[Repository \(documentation & codebook\)](#)

<https://github.com/owid/covid-19-data>

⚠ No key. The dataset was **frozen 19 Aug 2024** — scope the COVID overlay to 2019–2024.

OPTIONAL — DEEPER MACRO & POLICY DECOMPOSITION

4 · Oxford OxCGRT (COVID policy dataset)

Decompose the stringency index into containment vs. economic-support sub-indices.

[Repository — OxCGRT_compact_national_v1.csv](#)

<https://github.com/OxCGRT/covid-policy-dataset>

› No key; direct CSV in the repo.

5 · BLS Public Data API

Industry-specific payroll collapse / recovery (many of these series are also on FRED).

[BLS Developers — register for a v2 key](https://www.bls.gov/developers/)

<https://www.bls.gov/developers/>

› Free v2 key → 500 queries/day.

6 • BEA GDP-by-Industry

“Markets lead the real economy” — compare industry value-added recovery vs. ETF recovery.

[Request a free UserID \(36-char\)](https://apps.bea.gov/api/signup/)

<https://apps.bea.gov/api/signup/>

[BEA API documentation](https://apps.bea.gov/api/)

<https://apps.bea.gov/api/>

› Use DataSetName=GDPbyIndustry.